

Research Project Report

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Phase 1: Extract HOG Features

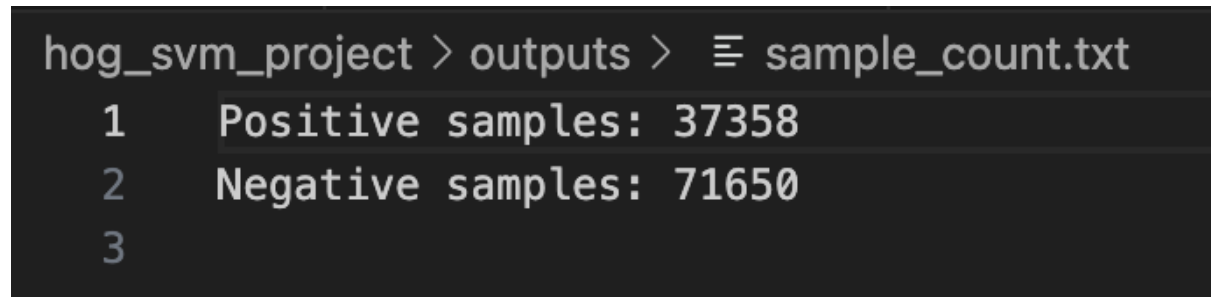
For human Dataset, we collect data from <http://mmlab.ie.cuhk.edu.hk/projects/PETA.html>, known as PETA dataset. It consists of 19000 images of pedestrians. Those images include 8705 persons with different recognizing attributes, such as gender and clothing style. Those images are with resolutions ranging from 17-by-39 to 169-by-365 pixels. Our model will use images of 64×128 pixels as input. The ratios of images from this dataset are approximately 1:2. Because of that, the resize on this dataset will be easy and won't cause image distortion. For better training effect, the images in training sets all contain one single person only. While in testing set for human, we select a variety of images, including crowds of people, partially occluding people and people in distance.

For non-human Dataset, we managed to find sets of background images from <https://github.com/koishi70/Landscape-Dataset>. It consists of 7268 images with pure landscape. The advantage of this dataset is it doesn't have any object from another class such as an animal or a vehicle. They're all typical background images without any objects in them. It is good for our training as our model won't be misled by human-like objects such as monkeys. After our selection, we sorted all our dataset into 4 documents: human training set, non-human training set, human testing set and non-human testing set. The testing sets both contain exactly 100 images. Then all our images were renamed for better visualization. They were all stored as jpg in dataset-big.

In preprocess, we use multiple techniques to process our image. First, we need to make sure all our images are 64×128 pixels, as our model will need input of the same quality for training. For our human dataset, all our training images are a human figure, resizing it directly to 64×128 pixels won't have too much impact on its quality. However, for our non-human dataset, all the images are 320×180 pixels. Resizing them directly could result in severe image distortion. Thus, we use `random_crop` to randomly cut the images with the ratio of 1:2. By using these methods, our negative samples will fit as input for our model. For each non-human image, we got 10 negative samples by `random_crop`. For human dataset we use

the method `mirror_resize` to generate its left-right version. Those two methods were inspired by the original essay, aiming to create more samples to train our model.

As Figure.1 shows, now we have 37358 positive samples and 71650 negative samples after preprocessing. All of them were stored in a directory: `data_preprocessed`.

A terminal window with a dark background and light gray text. The prompt is 'hog_svm_project > outputs > ≡ sample_count.txt'. The output consists of three lines: '1 Positive samples: 37358', '2 Negative samples: 71650', and '3' on the next line.

```
hog_svm_project > outputs > ≡ sample_count.txt
1 Positive samples: 37358
2 Negative samples: 71650
3
```

Figure.1 The output of training set sample count

Phase 2: Feature Extraction & Model Training:

All training and testing images are converted to grayscale with fixed resolution of 64×128 . The main idea of HOG is to represent local appearance by encoding the distribution of intensity gradients and edge orientations within small regions. We apply ‘`skimage.hog`’ implementation to extract features. Each detection window is 16×16 pixels—dividing into cells of 8×8 pixels and blocks of 2×2 cells. Cells are the smallest units for gradient histogram computing and blocks are units for normalization. Gradient orientations are aggregated into histograms, and the HOG feature vector is formed by concatenating the normalized orientation histograms from all blocks across the detection window. The default number of orientation bins is 9. We apply L2-Hys normalization scheme to enhance the robustness against illumination and contrast.

To train the basic classifier, feature vectors from all positive and negative patches are extracted and used to train a ‘`LinearSVC`’ model in ‘`scikit-learn`’ with `max_iter=10000`. The label 0 stands for non-human images and label 1 stands for human images. Inspired by Dalal and Triggs, we also use the “hard example” and “retrain” methodologies mentioned in their essay. After the first version of the model is ready, a slide window is used to sample exhaustively all the original images before a random sample mentioned in Phase 1. If any patch was falsely classified, the patch will be added to the negative training set. A new training set with “hard example” is constructed after the process. Then, the new set is applied to retrain a model as the same training process about. This process significantly improves the

model by mitigating the probability of reporting non-human images as human images. Figure.2 illustrates that the curve of the model after retrain is closer to the original point regarding any value of threshold. Obviously, the model is optimized by retrain significantly.

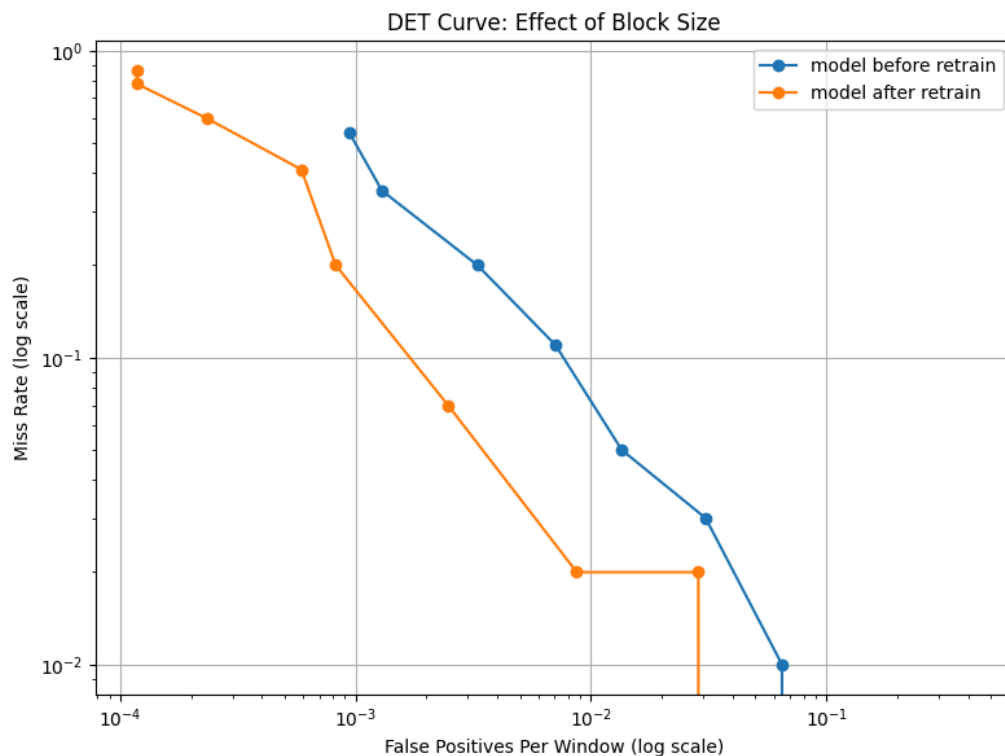


Figure.2 The performance of model before and after retrain on testing set 200 images

To assess the effectiveness of the final retrained model, we evaluated its performance using both image-level and window-level metrics. On the test set containing 100 positive and 100 negative images, The result of assessment shown in Figure.3.

```

===== Evaluation Report =====
[Image-level] Miss Rate: 0.0700
[Window-level] FPPW: 0.002471
[Image-level] Accuracy: 0.9100
[Image-level] Precision: 0.8942
[Image-level] Recall: 0.9300

```

Figure.3 The evaluation metrics of final model tested on testing set of 200 images

Miss Rate is the Proportion of human images where no window is classified as positive. The second metric indicates the average number of false detections per sliding window in non-human images. The accuracy is the percentage of correctly classified images among the entire testing set. Precision indicates how many images detected to be “human” are true “human” images. Finally, the recall means proportion of detected “human” images against all the “human” images, demonstrating how many true targets missed.

Phase 3: Ablation study

We selected three parameters for the ablation study: number of orientation bins, block size, and normalization method. The default values for these parameters are 9, 16*16, and L2-Hys, respectively.

Our approach is to vary only one parameter at a time relative to the default configuration, then train and retrain the model with this new setting, and obtain the corresponding final model. In addition to the default model, we create two more models for each parameter. We then evaluate the models for each parameter individually and generate three corresponding curves of FPPW (False Positives Per Window) and miss rate. These curves allow us to compare different values of each parameter to determine which performs best.

In each curve, the vertical axis represents the miss rate and the horizontal axis represents the FPPW, both plotted on a logarithmic scale. We vary the detection threshold across 11 evenly spaced values in the range of $[-1.5, 1.5]$. For each threshold, we compute the corresponding miss rate and FPPW, then connect these points to form a curve. Since lower values of both miss rate and FPPW are better, the closer a curve is to the lower left corner, the better the performance of that parameter setting. The test set contains 100 images of humans and 100 images of non-human objects.

For the number of orientation bins, in addition to the default value of 9, we also tested values of 6 and 12. The results are shown in the following figure:

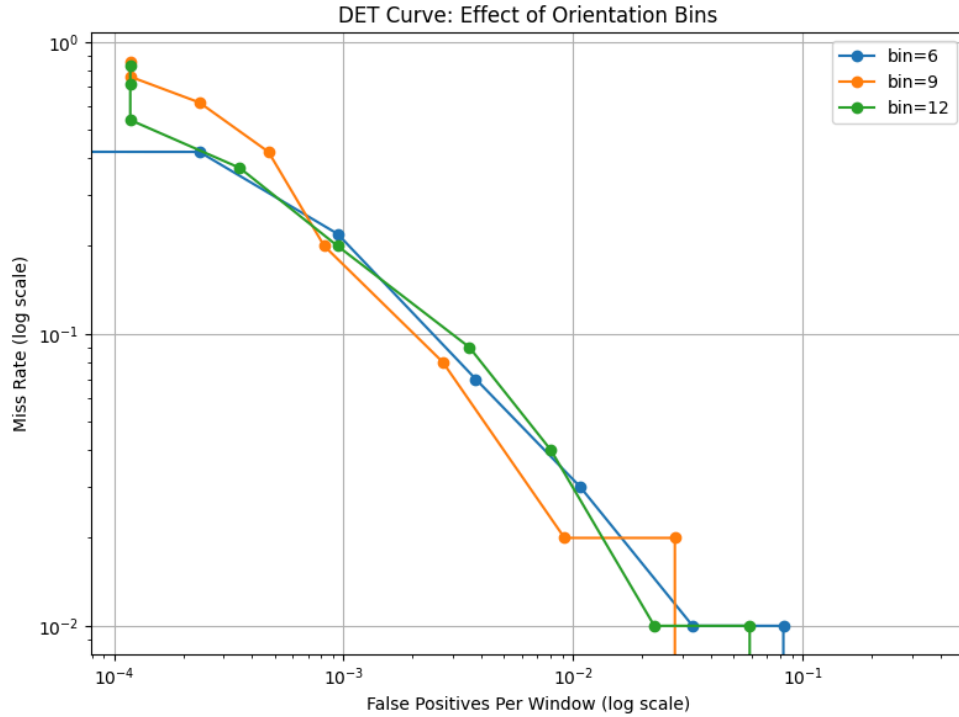


Fig. 4 DET Curve: Effect of Orientation Bins.

From the figure, we can observe that when the number of orientation bins is 9, the curve is generally closer to the lower left corner across most regions, indicating better overall performance. When the value is 12, performance is slightly worse than 9 at higher FPPW levels (right side). When set to 6, performance is inferior to 12 at lower FPPW levels (left side), and worse than 9 at higher FPPW levels, making it the least favorable overall.

For block size, in addition to the default value of 16*16, we also tested 8*8 and 32*32. The results are shown below:

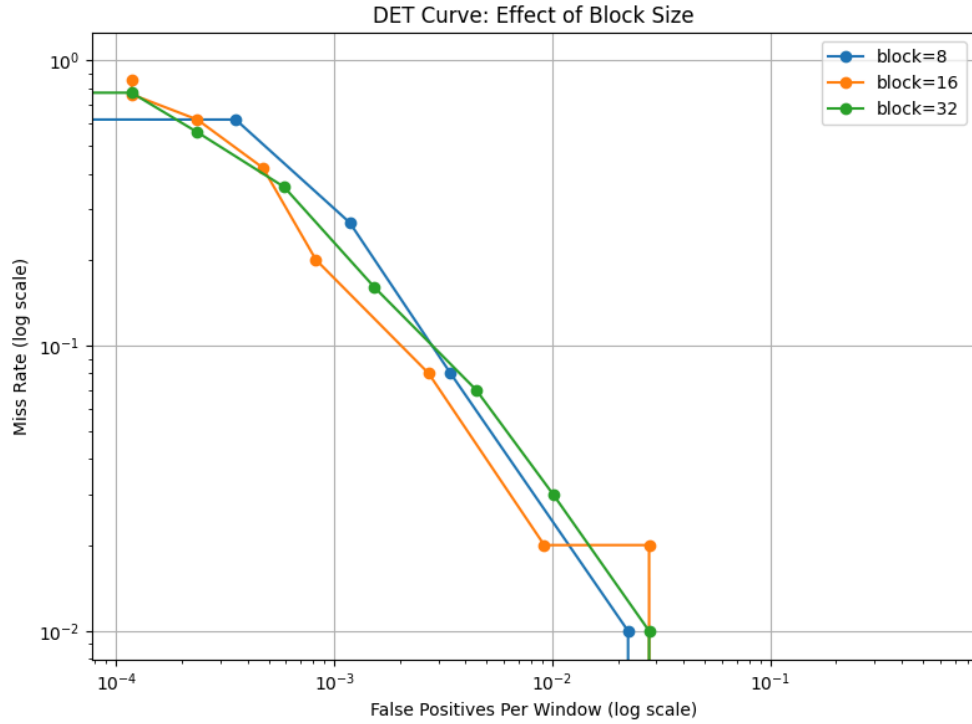


Fig. 5 DET Curve: Effect of Block Size

We can see that when the block size is 16×16 , the curve is generally closer to the lower left corner, suggesting the best overall performance. With a block size of 8×8 , the performance is consistently worse than 16×16 across most of the range. When using 32×32 , the performance at lower FPPW levels is comparable to 16×16 , but slightly worse at higher FPPW levels.

For normalization, besides the default method L2-Hys, we also tested L1 and L2. The results are as follows:

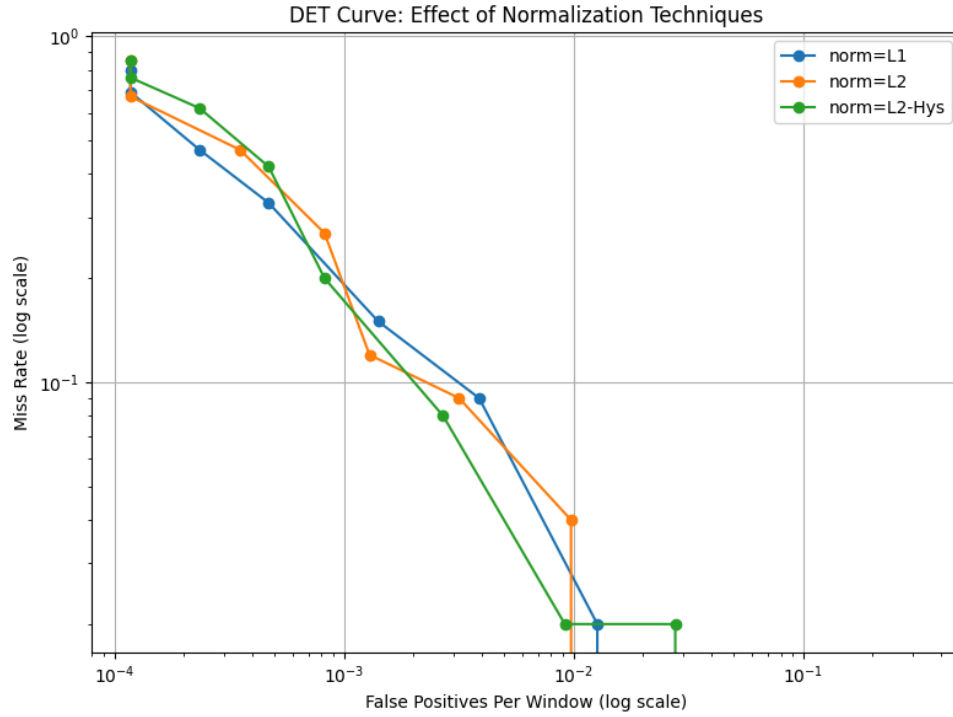


Fig. 6 DET Curve: Effect of Normalization Techniques

It can be seen that when using L2-Hys normalization, the curve remains closer to the lower left corner in most regions. The curves for L1 and L2 are relatively close to each other but both underperform compared to L2-Hys.

In summary, the model achieves the best performance when the number of orientation bins is set to 9, the block size is 16*16, and the normalization method is L2-Hys.

References

- [1] N. Dalal and B. Triggs, "Histograms of Oriented Gradients for Human Detection," CVPR, 2005.